

Exploratory Data Analysis (EDA) - Global Superstore 2016

Project Overview

This project is part of the CodeAlpha Data Analytics Internship.

The goal of this project is to perform Exploratory Data Analysis (EDA) on the “Global Superstore 2016” dataset using Python. The analysis explores sales, profit, customer segments, regional performance, and the relationship between discounts and profitability through data visualization.

Objectives

1. Understand the structure of the dataset.
2. Clean and inspect the data.
3. Check for missing values and duplicate records.
4. Analyze sales and profit across different categories.
5. Compare sales performance across regions.
6. Study customer segment contributions.
7. Analyze profit by product sub-category.
8. Explore the relationship between discounts and profit.
9. Generate meaningful business insights.

Technologies Used

1. Python 3
2. Pandas
3. Matplotlib
4. Seaborn
5. OpenPyXL
6. Visual Studio Code

Dataset

Dataset: Global Superstore 2016

The dataset contains information about:

1. Orders
2. Customers
3. Products
4. Sales
5. Profit
6. Discounts
7. Regions
8. Customer Segments

Project Structure

1. CodeAlpha_EDA
2. global_superstore_2016.xlsx
3. EDA.py
4. README.md
5. EDA_Report.pdf
6. Sales by category.png
7. Profit by category.png
8. Sales by region.png
9. Sales by segment.png
10. Profit by subcategory.png
11. Discount vs profit.png

Exploratory Data Analysis

The following analyses were performed:

1. Dataset Overview
2. Missing Value Analysis
3. Duplicate Record Analysis
4. Summary Statistics
5. Sales by Category
6. Profit by Category
7. Sales by Region
8. Sales by Customer Segment
9. Profit by Sub-Category
10. Discount vs Profit Analysis

Visualizations

The project includes the following charts:

1. Sales by Category
2. Profit by Category
3. Sales by Region
4. Sales by Customer Segment
5. Profit by Sub-Category
6. Discount vs Profit (Scatter Plot)
- 7.

Key Insights

1. Sales performance varies across different product categories.
2. Profit differs significantly between categories and sub-categories.
3. Customer segments contribute differently to overall sales.
4. Regional sales performance is not uniform.
5. Some product sub-categories generate low or negative profits.
6. Higher discounts often lead to reduced profitability.

Skills Demonstrated

1. Data Cleaning
2. Data Exploration
3. Data Visualization
4. GroupBy Operations
5. Business Insight Generation
6. Python Programming
7. Exploratory Data Analysis (EDA)

Future Improvements

1. Add an interactive dashboard using Power BI or Tableau.
2. Build interactive charts using Plotly.
3. Perform predictive analysis using Machine Learning.
4. Add automated report generation.

Author

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https://github.com/kapilrawat17/CodeAlpha_EDA.git